AWS VPC PEERING

Connecting Two Private Networks

PROJECT REPORT

Submitted by: Ayyan Yunus Attar

Programme: B.Tech – Computer Engineering

Institute: Eaglewood Polytechnic Institute

Internship: EWB EDU TECH – AWS & DevOps

AWS Region: Asia Pacific (Sydney) – ap-southeast-2

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1. Project Overview

This project demonstrates communication between two separate Amazon Virtual Private Clouds (VPCs) using VPC Peering. The peering connection links the private network ranges so resources in the configured subnets can communicate using private IP addresses.

2. Objectives

• Create two isolated VPC networks with non-overlapping CIDR ranges.

• Establish and activate a VPC Peering connection.

• Configure routes in both VPC private route tables.

• Configure security rules to permit the required test traffic.

• Verify private network connectivity using ICMP ping.

3. AWS Environment and Network Configuration

Resource	VPC A	VPC B
VPC name / ID	ewb-vpc-a vpc-0f1dce2afef60b65c	ewb-vpc-b vpc-04ea32ba169eb3bad
IPv4 CIDR	10.10.0.0/16	10.20.0.0/16
Private subnet	ewb-subnet-a-private 10.10.2.0/24	ewb-subnet-b-private 10.20.1.0/24
Private EC2 instance	ewb-app-a 10.10.2.10	ewb-app-b 10.20.1.10

4. VPC Peering Connection

Peering connection name: ewb-a-b-peer
Peering connection ID: pcx-0f8becfc5a5c6ada4
Status: Active
The peering connection was created between VPC A (requester) and VPC B (accepter) in the same AWS Region.

5. Route Table Configuration

The private route table in each VPC contains a local route for its own CIDR and a route to the peer VPC CIDR through the active peering connection.

• VPC A private route table (ewb-rt-a-private): 10.10.0.0/16 → local; 10.20.0.0/16 → peering connection (Active).

• VPC B private route table (ewb-rt-b-private): 10.20.0.0/16 → local; 10.10.0.0/16 → peering connection (Active).

• VPC A public route table also shows a route to 10.20.0.0/16 through peering; its default route points to the Internet Gateway.

6. Compute and Security Configuration

The lab uses a bastion instance in the public subnet of VPC A and two private application instances:

• Bastion: ewb-bastion – public IP 3.26.16.2, private IP 10.10.1.10.

• App-A: ewb-app-a – private IP 10.10.2.10.

• App-B: ewb-app-b – private IP 10.20.1.10.

The App-B security group includes an SSH inbound rule from the bastion private IP (10.10.1.10/32) and an ICMP inbound rule from App-A (10.10.2.10/32).

7. Connectivity Testing and Results

From App-A, the command `ping -c 4 10.20.1.10` was executed to test connectivity to App-B. The captured output shows four replies, four packets received, and 0% packet loss. The average round-trip time shown in the screenshot is 0.172 ms.

SSH limitation: The captured TCP port 22 check returned CLOSED. Therefore, SSH connectivity from App-A to App-B is not confirmed and is not claimed as successful in this report.

8. Evidence Screenshots

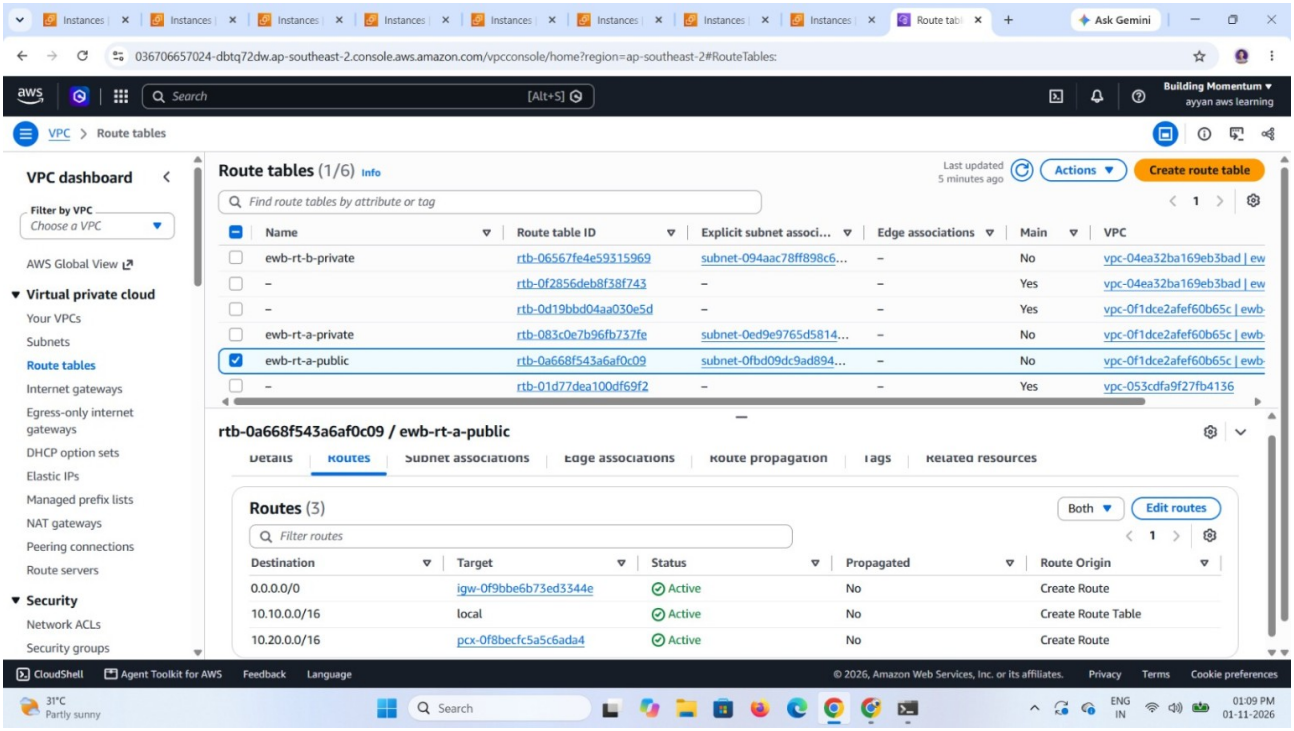


Figure 1. Successful connectivity test from App-A (10.10.2.10) to App-B (10.20.1.10): 4 packets transmitted, 4 received, 0% packet loss.

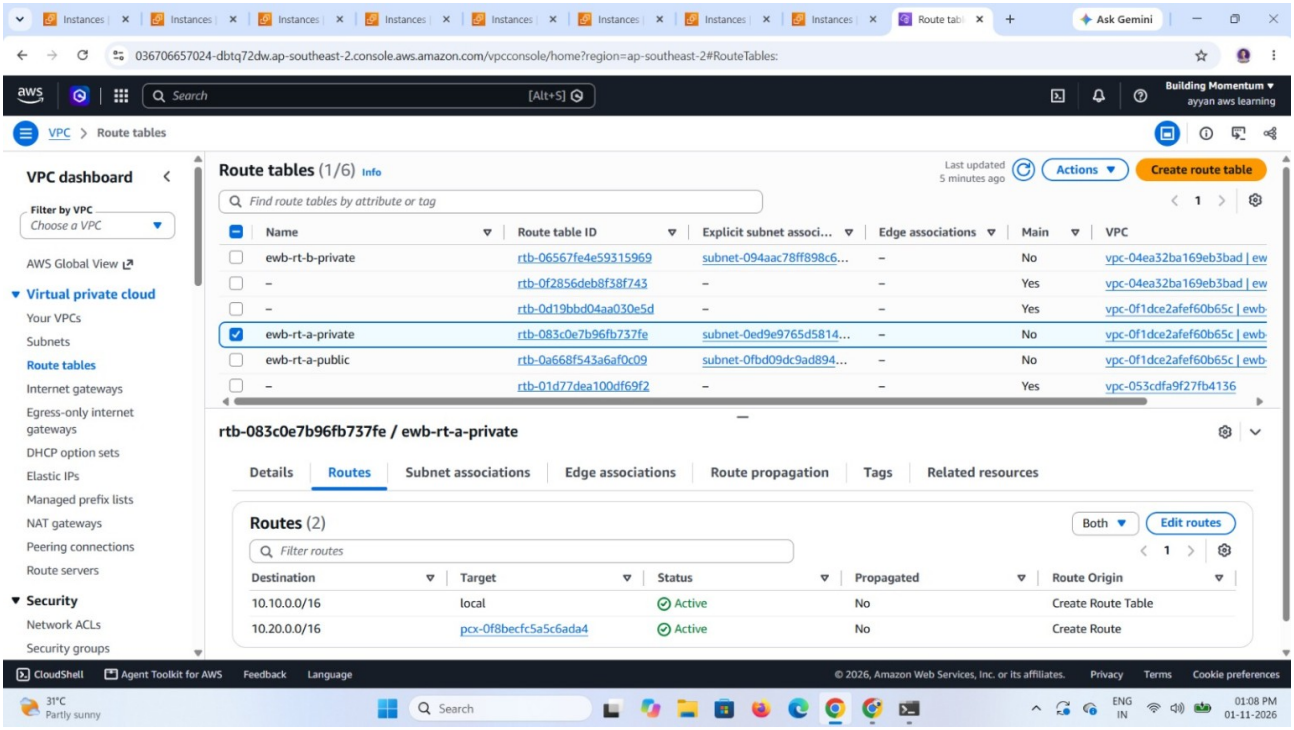


Figure 2. VPC A private route table: local route and active peering route to 10.20.0.0/16.

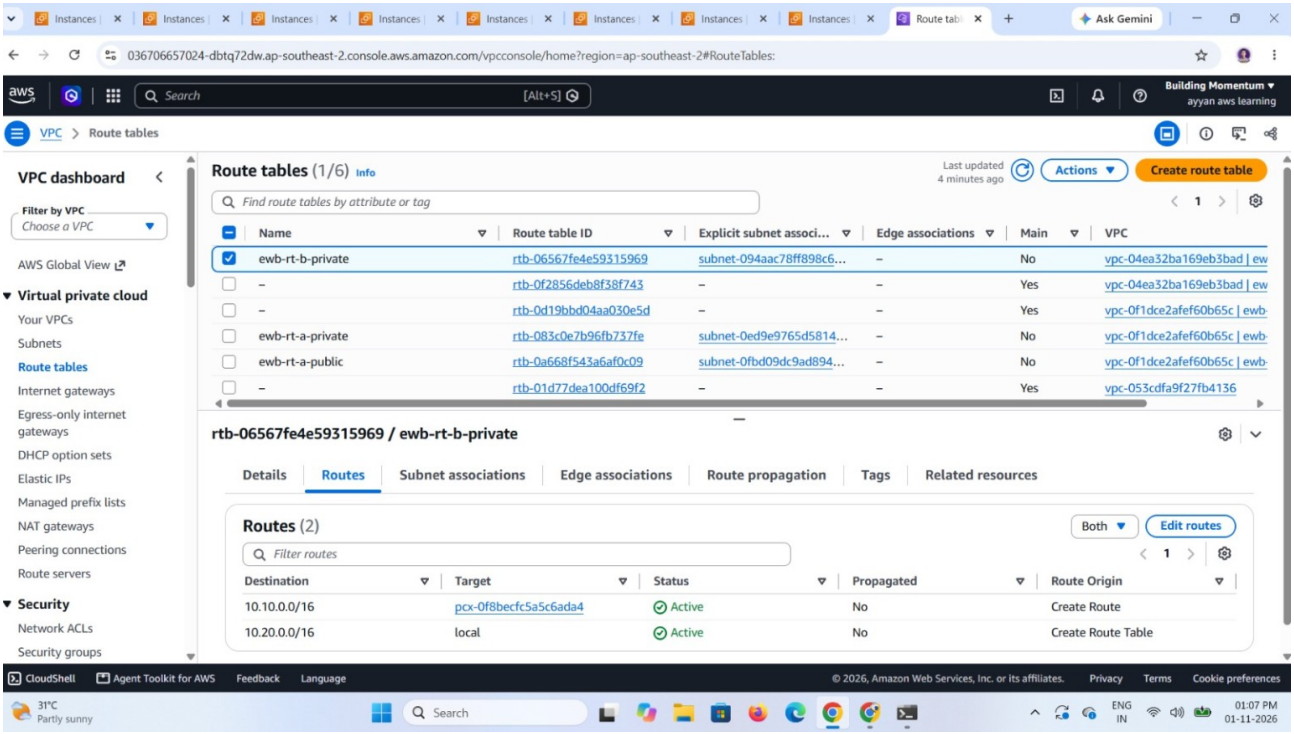


Figure 3. VPC B private route table: active peering route to 10.10.0.0/16 and local route.

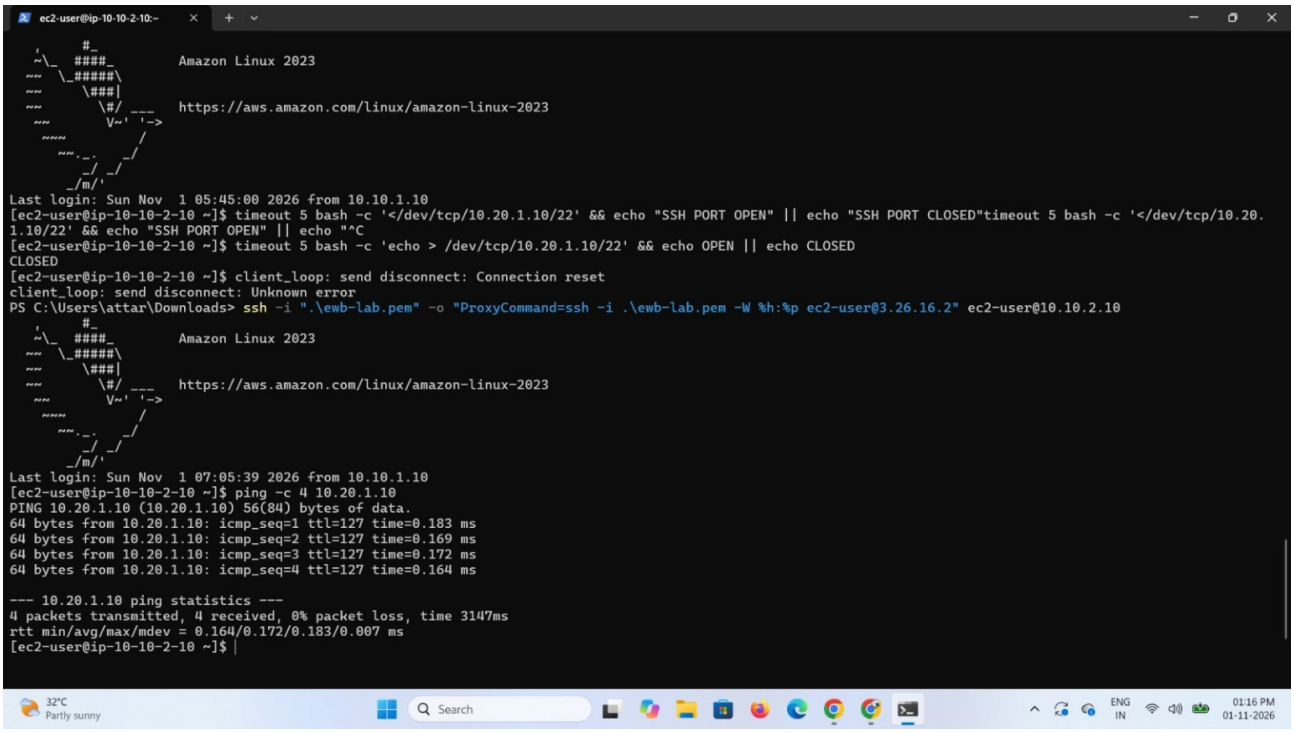


Figure 4. Terminal evidence: SSH port 22 check returned CLOSED; the ICMP ping test below it succeeded.

9. Conclusion

The VPC Peering connection between ewb-vpc-a and ewb-vpc-b is active, and the private route tables contain the required routes to the peer CIDR ranges. The successful ICMP test from App-A to App-B confirms private IP connectivity between the two VPC networks for the tested traffic. SSH access was not verified, as the port 22 check returned CLOSED.

10. Evidence Checklist

• Active VPC Peering connection

• VPC A private route table

- VPC B private route table
- Successful App-A to App-B ping output
- SSH limitation accurately documented